

Introduction to Vector & Co-ordinate system

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

Dot product

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$\text{Projection of } \vec{A} \text{ on } \vec{B} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|}$$

Cross product

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta \hat{n}$$

$$\sin \theta = \frac{|\vec{A} \times \vec{B}|}{|\vec{A}| |\vec{B}|}$$

Newton's law of Motion

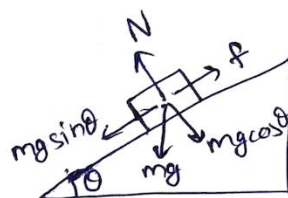
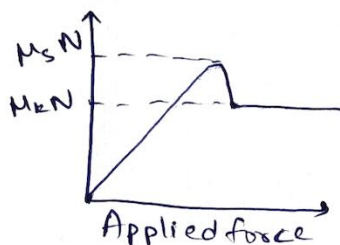
$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

$$p = mv$$

$$f_s < f_s^{\max} = \mu_s N$$

$$f_s = \mu_s N$$

$$f_k = \mu_k N$$



$$\mu = \tan \theta$$

$$a = g(\sin \theta - \mu_k \cos \theta)$$

$$\mu_s = \tan \theta_c$$

$\theta_c = \text{angle of } \cancel{\text{transport}} \text{ repose}$

Momentum & Impulse

Linear momentum (P)

$$p = mv$$

$$\Sigma \vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{d\vec{p}}{dt}$$

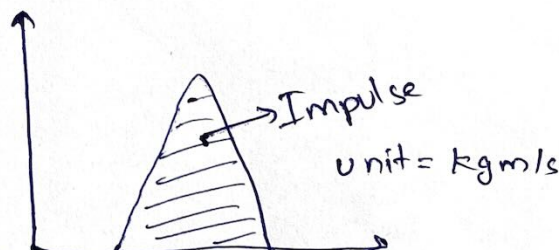
$$\int_{t=t_i}^{t=t_f} \vec{F}(t) dt = \vec{I}$$

$$F = \frac{d\vec{p}}{dt} = \frac{p_f - p_i}{t_f - t_i}$$

$$d\vec{p} = F dt$$

$$I = F \Delta t$$

$$I = p_f - p_i$$



Conservation of momentum

If no net external force acts on a system

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

Work & kinetic energy

$$KE = \frac{1}{2} mv^2 = \frac{p^2}{2m}$$

$$m_1 > m_2$$

$$KE_1 < KE_2$$

$$W = \vec{F} \cdot \vec{s} = FS \cos \theta = \int_A^B \vec{F} \cdot d\vec{s}$$

Work energy theorem

$$W_{\text{total}} = K_f - K_i = \Delta K \quad (\text{Prove using third law of motion})$$

Conservation of Mechanical energy

Conservative force

$$W_1 = W_2 = W_n$$

Non-Conservative force

$$W_1 \neq W_2 \neq W_n$$

Conservation of energy

$$\Delta E = \Delta E_{\text{system}} + \Delta E_{\text{surrounding}} = 0$$

Force & PE

$$\vec{F} = - \left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right)$$

Restoring force

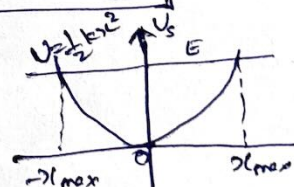
$$\vec{F} = - \frac{dU}{dx} = -kx \hat{e} \rightarrow \text{Hooke's law}$$

$$\Delta U = -W = - \int \vec{F} \cdot d\vec{s} = \frac{kx^2}{2}$$

Conservation of ME

$$U_i + K_i = U_f + K_f$$

$$PE = mg(h_2 - h_1)$$



Optics UNIT-II

→ Wave Propagation :

$$y_1 = A \sin(kx \pm \omega t) = A \sin\left(2\pi \left[\frac{x}{\lambda} \pm \frac{t}{T}\right]\right)$$

$$\boxed{k = \frac{2\pi}{\lambda}} \quad \boxed{\omega = \frac{2\pi}{T}} \quad \boxed{f = \frac{1}{T}} \quad \boxed{v = \frac{\omega}{k} = \lambda f}$$

→ Superposition of sinusoidal waves :

$$\sin a + \sin b = 2 \cos\left(\frac{a-b}{2}\right) \sin\left(\frac{a+b}{2}\right)$$

→ Path difference :

$$\delta = r_2 - r_1 = d \sin \theta$$

→ Condition for constructive interference or Bright fringe

$$\delta = d \sin \theta_{\text{Bright}} = m\lambda, \quad (m = 0, \pm 1, \pm 2, \dots)$$

→ Condition for destructive interference or DARK fringe

$$\delta = d \sin \theta_{\text{Dark}} = \left(m + \frac{1}{2}\right)\lambda, \quad (m = 0, \pm 1, \pm 2, \dots)$$

→ the linear positions of bright are given by:

$$y_{\text{Bright}} = \frac{\lambda L}{d} m$$

$$\& \quad d \sin \theta_{\text{Dark}} = \left(m + \frac{1}{2}\right)\lambda, \quad (m = 0, \pm 1, \pm 2, \dots)$$

→ the linear positions of dark fringe are given by:

$$y_{\text{Dark}} = \frac{\lambda L}{d} \left(m + \frac{1}{2}\right)$$

→ Fraunhofer Diffraction

$$\boxed{\frac{a}{2} \sin \theta = \pm \frac{\lambda}{2}}$$

MODERN PHYSICS

→ Rayleigh - Jeans formula :

$$G(\nu) d\nu = \frac{8\pi \nu^2 d\nu}{c^3}$$

$$E_{avg} = kBT$$

Boltzman constant $\Rightarrow k_B = 1.381 \times 10^{-23} \text{ J/K}$

$$u(\nu) d\nu = E_{avg} \cdot G(\nu) d\nu = \frac{8\pi \nu^2 k_B T d\nu}{c^3}$$

→ Max Planck gave Radiation formula :

$$u(\nu) d\nu : \frac{8\pi h}{c^3} \cdot \frac{\nu^3 d\nu}{e^{h\nu/kT} - 1}$$

$$h = 6.626 \times 10^{-34} \text{ J.s}$$

if $h\nu \ll kT \Rightarrow e^{h\nu/kT} \rightarrow 1 + \frac{h\nu}{kT}$

→ Weins Displacement law :

$$\lambda_{peak} = \frac{b}{T}$$

$$b = 2.897 \times 10^{-3} \text{ mK}$$

→ Photo Electric Effect :

$$\text{Max Kinetic Energy} = K_{max} = (-e)(-\Delta V) = e\Delta V_s.$$

→ Quantum Theory of light:

$$E = h\nu = \frac{hc}{\lambda}$$

→ Photo Electron Energy:

$$KE_{max} = h\nu - \phi = h\nu - h\nu_0$$

$$KE_{max} = h(\nu - \nu_0)$$

$$h\nu = \phi + KE_{max}$$

$$KE_{max} = \frac{1}{2}mv^2$$

$$h\nu = \phi + \frac{1}{2}mv^2$$

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→ Max Kinetic Energy (KE_{max})

$$h\nu = \phi + \frac{1}{2}mv^2$$

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→ Photo Displacement Law:

$$KE_{max} = \frac{1}{2}mv^2$$

→ Photo Electric Effect:

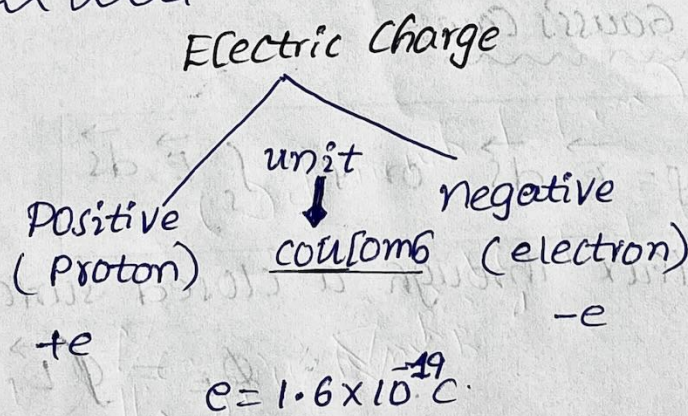
$$KE_{max} = h\nu - \phi$$

* UNIT-4 " Formulae sheet *

Electromagnetism - 1

1. → Electrostatic field
2. → Flux and Gauss's law
3. → Application of Gauss's law due to infinite wire and sheet.
4. → Electrostatic field due to conducting and insulating sphere.
5. → Electrostatic potential and Potential energy
6. → Capacitor and capacitance

1. Electrostatic field :-



* Charge is quantized : N number of electrons have charge $-Ne$
N number of protons have charge $+Ne$

Like charges repel,

unlike charges attract.

Coulomb's Law :-

$$F = K \frac{q_1 q_2}{r^2}$$

The force acts along the line joining the two.

$$K = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$$

ϵ_0 is Permittivity of free space

$$= 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$$

Electric field :-

$$E = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} \right)$$

q is the source charge

For several point charges $q_1, q_2, \dots, q_n$ at distances

$r_1, r_2, \dots, r_n$ from q .

$$\vec{E}_{\text{total}} = \vec{E}_1 + \vec{E}_2 + \dots = \sum_{i=1}^N \vec{E}_i$$

$$E = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i$$

2. Flux and Gauss's Law :-

Flux :-

$$\Delta\phi = \vec{E} \cdot d\vec{s} \quad \text{or} \quad \phi = \int_s \vec{E} \cdot d\vec{s}$$

Electrostatic flux through a closed surface.

$$\phi_E = \oint \vec{E} \cdot d\vec{A}$$

Net flux
over closed
surface

Integral
over closed
surface

Gauss's law :-

$$\phi = \oint_s \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

A vector representing an infinite small element of area of surface.

3. Application of Gauss's Law: E field due to infinite wire and sheet:-

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

symmetry	Gaussian surface
spherical	concentric sphere
cylindrical	coaxial cylinder
planar	Gaussian "pillbox"

- when the charge distributed continuously in some region, the sum becomes.

$$E = \frac{F}{Q} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i \xrightarrow{\text{integral}} E(r) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{r} dq$$

Continuous charge Distribution:-

infinitesimal charge $dq = \lambda dl = \sigma da \equiv \rho dv$

$\lambda \Rightarrow$ linear charge density [C/m]. 

$\sigma \Rightarrow$ surface charge density [C/m²]. 

$\rho \Rightarrow$ volume charge density [C/m³]. 

Electrostatic field due to infinite wire:

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \hat{r}$$

Electrostatic field due to infinite sheet:

$$E = \frac{\sigma}{2\epsilon_0}$$

4. Electrostatic field due to conducting and insulating sphere:-

Insulating sphere:-

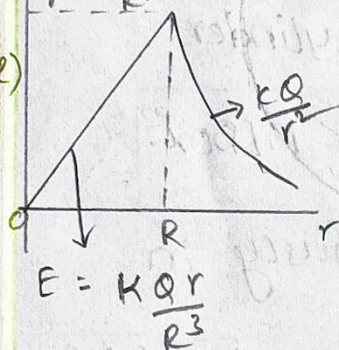


$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2} \text{ for } r > R$$

$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 R^2} \text{ for } r = R$$

$$\Rightarrow E = \frac{Qr}{4\pi\epsilon_0 R^3} \text{ for } r < R$$

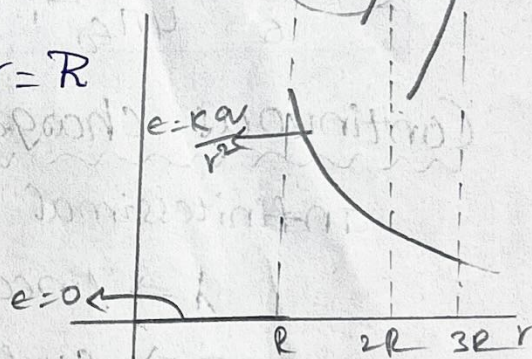
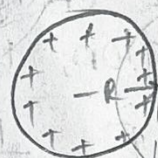
$E(r)$



Conducting sphere:- $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \text{ } r > R$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \text{ } r = R$$

$$\Rightarrow E = 0 \text{ } r < R$$



5. Electrostatic Potential and potential energy:-

Electric Potential energy:-

where,

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

r = distance b/w two charges

• for a system of charges $q_1, q_2, q_3 \dots$ etc

$$U = \sum_{i < j} \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}}$$

work done

$$W_{a \rightarrow b} = U_a - U_b = -\Delta U$$

Electric Potential:-

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

where, r is the distance b/w point charge and the Point of potential we need to measure at.

- Electric potential due to continuous distribution of charges $\Rightarrow V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$

- Electric field and potential are closely related by

$$V(r) = - \int_{\infty}^r \vec{E} \cdot d\vec{r}$$

- Electric field vector from potential:

$$\vec{E} = - \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) V = - \vec{\nabla} V = - \text{Grad } V$$

Trick
Electric potential due to a sphere (conducting or insulating).

Potential at outside: No need to break the integral, do the integration.

Potential at inside: Break integral into two parts. ∞ to R , and R to r .

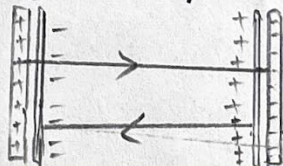
6. Capacitor and Capacitance:-

~~C = Q/V~~

$$C = \frac{Q}{V} \rightarrow \text{unit: Faraday}$$

capacitance due to parallel plate capacitor.

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$



UNIT - V Electromagnetism

* Magnetic force on a moving charge

$$\vec{F}_B = q\vec{V} \times \vec{B} \iff F_B = qVB \sin \theta$$

Unit (T) Tesla ; $1T = 10^4 \text{ Gauss (G)}$

Here ; ' θ ' is the angle between $\vec{V}$ and $\vec{B}$.

* Lorentz Force

$$\begin{array}{l} \vec{F}_E = q\vec{E} \\ \text{\{Electric Fields\}} \end{array} \quad \begin{array}{l} \vec{F}_B = q\vec{V} \times \vec{B} \\ \text{\{Magnetic fields\}} \end{array}$$

$$\boxed{\vec{F} = q(\vec{V} \times \vec{B} + \vec{E})}$$

* Biot-Savart's law

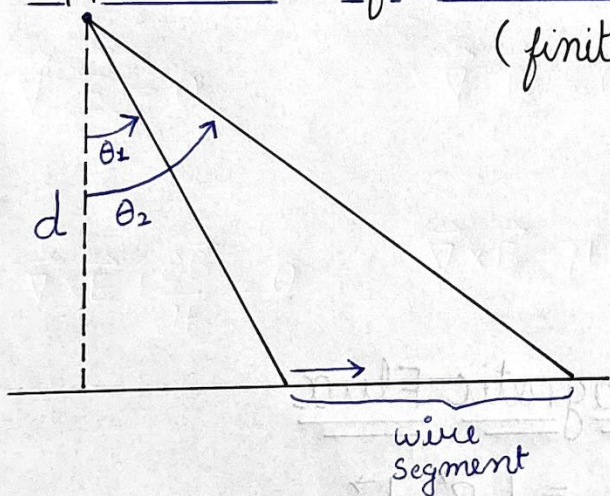
μ_0 = permeability of Vacuum
 $\rightarrow 4\pi \times 10^{-7} \text{ Tm A}^{-1}$

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl \sin \theta}{r^2} ;$$

$\Rightarrow$ In Vector form;

$$\vec{B} = \int d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2}$$

* Application of Biot-Savart wire (finite wire)



$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_1 - \sin \theta_2)$$

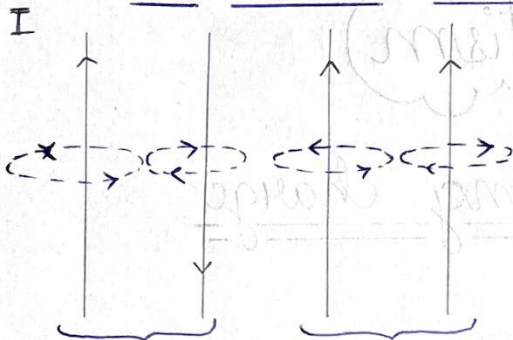
* field due to long wire
 (infinite wire)

* Magnetic force on current carrying wire:

$$\vec{F}_B = I(\vec{L} \times \vec{B})$$

$$B = \frac{\mu_0 I}{2\pi d} \hat{\phi}$$

* Two Parallel Wires



* here Magnetic force adds up * here Magnetic force cancels

* Current Carrying Wire

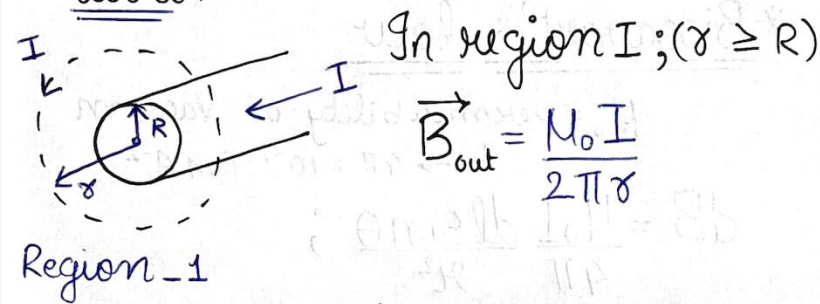
$$f = \frac{dF}{dl} = \left(\frac{\mu_0 I_1 I_2}{2\pi d} \right)$$

* Ampere's Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

$$I_{enc} = \int_S \vec{J} \cdot d\vec{A}$$

* Magnetic field due to infinite wire:-

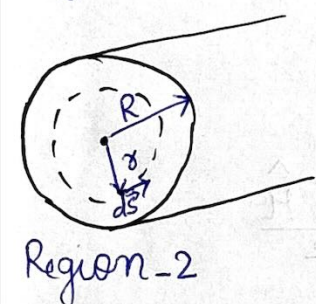
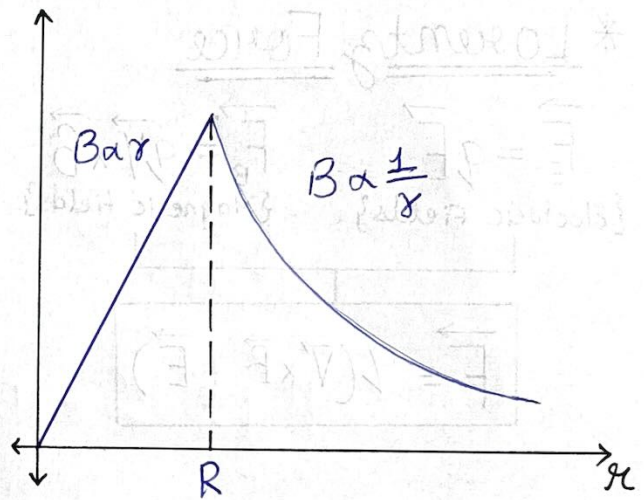


In region I; ($r \geq R$)

$$\vec{B}_{out} = \frac{\mu_0 I}{2\pi r}$$

In region II; ($r < R$)

$$\vec{B}_{in} = \frac{\mu_0 I r}{2\pi R^2}$$



Region-2

* Magnetic field due to Solenoid

$$B = \mu_0 n I$$

• n represents no. of turns per unit length

* Faraday's Law

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

'-' Negative sign in the equation is due to Lenz's Law.

* Magnetic Flux

$$\Phi_B = \int \vec{B} \cdot d\vec{s}$$

• Uniform field $\Rightarrow \Phi_B = BA \cos \theta$

• SI Unit $\Rightarrow T \cdot m^2 \equiv \text{Weber}$

* Gauss's Law of Magnetism * Displacement Current

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$I_d = \epsilon_0 \frac{d\phi_E}{dt}$$

* Ampere-Maxwell Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

* Integral form of Maxwell's relation are:

$$1. \oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int_V \rho \, dv \quad (\text{Gauss law of Electronics})$$

$$2. \oint \vec{B} \cdot d\vec{s} = 0 \quad (\text{Gauss law of Magnetostatics})$$

$$3. \oint \vec{E} \cdot d\vec{l} = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} \quad (\text{Faraday's law of EMI})$$

$$4. \oint \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{J} \cdot d\vec{s} + \mu_0 \epsilon_0 \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{s} \quad (\text{Ampere's law modified by Maxwell})$$

* Maxwell's Equation in Free Space:

$$\begin{array}{ccc} \vec{\nabla} \cdot \vec{E} = 0 & \vec{\nabla} \cdot \vec{B} = 0 & \rho = 0 \\ & & \vec{J} = 0 \\ \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 & \vec{\nabla} \times \vec{B} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = 0 & \end{array}$$

* Maxwell's equation (Differential form)

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0, \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}, \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$